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Ravdness, Crema-D, Tess based algorithm for Emotion Recognition Using Speech

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Abstract— Speech is the most important communication among people. While interfacing, the emotions of others can be easily comprehended. Nonetheless, this isn't the situation when a human and a machine connect. In this work, machines are made to identify and understand emotions of the human while interacting. The decisions people make are mostly based upon the emotions in many scenarios. The investigations of feeling extraction from human voice are still new and draw in extraordinary consideration. The essential difficulties of feeling acknowledgment are picking the feeling acknowledgment corpora (speech dataset), recognizable proof of various highlights identified with discourse and a suitable decision of an arrangement model. Extracting the required data for emotion extraction poses an important problem, In the modern era of artificial intelligence perception of human emotions the objective of these research paper is to overcome the emotion detection. CNN (Convolution Neural Network) models have been implemented on CREMA-D, RAVDNESS, TESS datasets by adding additional speckel noise to the datasets.

Keywords— Convolution Neural Network, Emotion, classification, CREMA-D, RAVDNESS, TESS

I. INTRODUCTION

Human machine interaction is widely used nowadays in many applications [15]. In day to day lives, human beings are always dependent on communication which may be of any form like hand-signs, facial expressions, vocal etc. Image and voices were the most common features which help in integration with machines make easier to overcome the needs of automation. Discourse is the most proficient and compelling way for correspondence between people.[18].

Emotions are inherent in human nature. To understand people's emotions, intelligent systems based on multi-modal emotion detection are increasingly emerging [8]. People express their emotions while speaking in different ways, i.e, it may not be same for any two individual and moreover it may vary when a person is tested under extreme circumstances. A major difficultly for speech emotion recognition is the reduction of noise which incorporated with the speech data sample [15]. The commonly used features in speech recognition are MFCC (Mel Frequency spectrum Coefficient), prosodic features, sound quality characteristics and acoustic features [3]. Studies are being conducted to make machines work like humans by detecting the emotions of the humans in an accurate way which makes human machine interaction (so realistic that it makes the human to hallucinate in such a way that it's almost impossible to distinguish whether the person interacting on the other side is actual human or a machine) to a closer way. Extracting effective audio emotion features is a challenging task [8]. Existing emotion detection methods extract emotion features from audio signals directly, ignoring the diversity of content information, which makes the emotion recognition model less efficient [8]. These Emotions detection system helps in many organizations in many ways such as to detect fake SOS messages, customer care call centers can be replaced, marketing, cognitive Science. It can also be used by to detect the mental situation of a person under critical circumstances.

Humans' emotions are exhibited by different persons in diverse ways expressing an emotion can be varied from one person to other person to linearize these all emotion, speech rate, pitch, energy value, tone and loudness are adjusted. To build an emotion recognition system it is very much important to extract different prosodic and acoustic features from speech signal [15]. To get to various kind of feelings with various varieties, diverse datasets are utilized. Among multi-modular feeling articulations and insights which comprises of various sorts of informational collections like CREMA-D, RAVDNESS, TESS, this paper mainly focus on CREMA-D, RAVDNESS, TESS datasets. The feelings like anger, happy, sad, fear, disgust and neutral are examined.

This paper addresses the execution of emotion recognition from voice datasets with a profound Convolutional Neural Network engineering (CNN) that process and classifies voice samples.[6] and using TensorFlow as backend. A convolutional neural network consists of an input layer, hidden layers and an output layer. In any feed-forward neural network, any middle layers are called hidden because their inputs and outputs are hidden by the activation functions and other parameters. Convnet works in analogous manner same as neurons in human brains. CNN decreases the error rate by 5-10% [3] which was superior to DNN. CNN LSTM network can also be used, but the effectiveness was found to be less [19].

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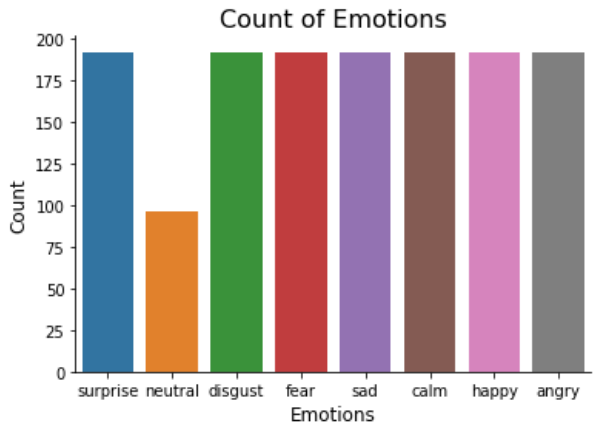

Fig:1.1: Count of Emotions

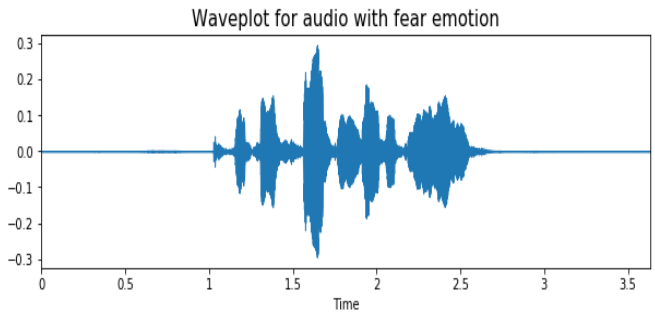

Fig:1.2: Waveplot for audio with fear emotion

II.Proposed Model

We are using CNN; A Convolutional Neural Network (CNN) is a sort of feed-ahead artificial network in which the joining sequence among its nodes is motivated by presenting an animal visual-cortex [13]. 3 types of datasets Ravdness, Crema-D, and Tess datasets are used.

A note of how the features are extracted:

1]Zero-Crossing Rate: - It's regarding the number of Times audio frame rate changes from positive to negative and other way around.

2]MFCC (Mel-Frequency cepstral Coefficients):-MFCC is utilized to Covert regular frequencies to Mel scaled frequencies.

3]Chroma STFT: - A chromogram is created from wave Spectrogram [6]

4]RMS:- Root Mean Square value calculates value for each frame of the data.

5]Mel Scale Spectrogram:- In this Spectrogram frequencies are converted into mel-scale frequencies.

The sample audio is provided as input along with the noise added to it. "Librosa" library whose main function is to analyze the audio and extract data from audio files is imported. In the next step, integers are assigned to the emotions and plotted a bar graph representing all available emotions in the dataset. Next, Waveplot and Spectrograph are plotted for the available emotions in the dataset and then noise is added to the dataset and again Waveplot and Spectrograph are plotted and compared with and without noise. Then at that point, the features are extracted from the dataset and the information is remixed and separated to train and test them. Then, they are passed to the CNN model and it's following layers to train the dataset and scale the values using Standard Scalar to make the values to our convenience. RELU is applied as the actuation work afterward toward the end Softmax is applied to standardize the result. To get better results, Reduce learning rate is applied (monitor = 'loss', factor = '0.4', verbose = 1, patience = 2, min_lr = 0.0000001). At last, the confusion matrix is plotted along with accuracy and classification report.

In this paper, a convolutional neural network is proposed to plot, among which one model was assembled utilizing 5 layer of neural network. Convolutional layers and max pooling are the key of this model. Layers include 3 dropout layers, 1 flatten layer, 2 fully connected (Dense Layer). Alternating convolution and max-pooling layers are composed of the upper-layers. The lower layers however are fully connected, flatten layers. To improve the accuracy of the model, some amount of noise is added to the data it means that the network is less able to identify the features of the training data because it causes some frequencies to changes and low pitch sound frequencies cannot be identified. Also stretched the data to increase the time by the fixed rate over a time. Shifted the to either right or left with a randomized value which helps in entire distribution of data points between the distances k, and the changing the pitch of the data which means adding a random change in pitch. Then used one hot encoding to classify multiple classes. Then splitted the data into training and testing datasets and by using StandardScaler standardized the data in between [0-1] and based on model adjusted the dimensions of the model. Then built a models using 3 layer kernel size with 5 convolution layers and 5 maxpooling layers and using activation function as RELU (Rectified Linear Unit).and a dense layer with 256 units with a drop out function rate as 0.3.

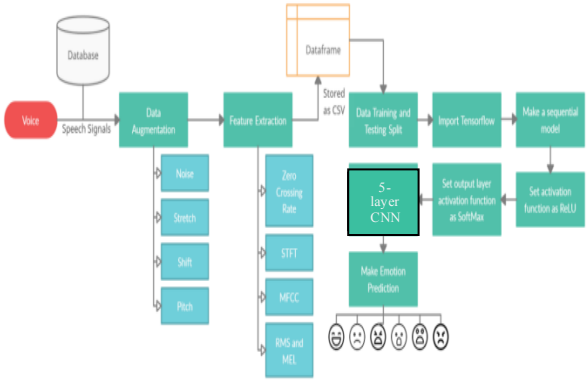

Fig:2.1:Data Flow Diagram

In this described model, a 5-layer CNN (Convolution neural networks) is used which consists of 5 convolution layers, 5 maxpooling layers, 3 dropout layers, 2 dense layers and 1 flatten layer. Each layer has its specific functionality.

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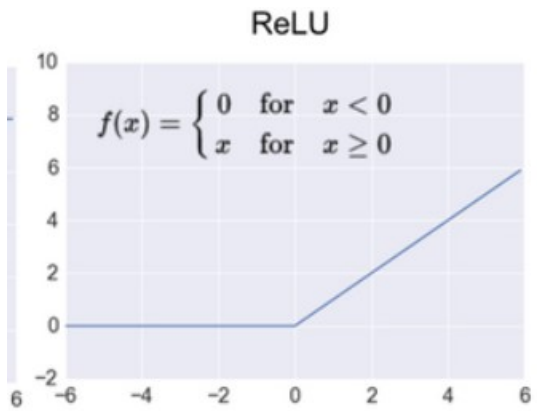

Fig: 2.2: ReLu Activation function

We had used activation function (Fig:2.1) as "Relu" The first layer(conv1d_1) it feed with 256 filters and then it was passed as an input to the maxpooling layer (max pooling 1) of pool size=3 then gets reduced to shape 53 and the shape was (None ,53,256) was given as an input to the second convolution layer(conv1d 2).This convolution layer(conv1d 2) reduces the 256 filters into 128 filters. Then the result was (None,51,128) and again given as an input to the second maxpooling layer(max poolingid 2). It consists of pool size=3 then it gets reduced from 51 to 17 and the shape of (None,17,128) was passed down to dropout layer(dropout 1) with a dropout function of 0.25 and then further it will be fed to another convolution layer which is (conv1d_3). The conv1d_3 reduces 128 filters into 64 filters and gets reduced to the shape(None,15,64), which was given as an input to third maxpooling layer(max poolingid 3) of pool size=3 then it gets reduced from 15 and the shape(None ,5,64) , this was given to second dropout layer(dropout 2) with a dropout function of 0.25 and then further was fed to fourth convolution layer(conv1d_4) which reduces the shape into (None ,3,64). and again, it was fed to another maxpooling layer(max poolingid 4) which does not affect the shape and number of filters the output of these layer was feed to fifth convolution layer(conv1d-5) which shape and filter into (None,1,32) and then given to fifth maxpooling layer(max_poolingid_5) which doesn't affect the features and shape. Then the layers are flattened which converts into [1,1]. Which was then fed to fully connected dense_1 layer output shape as (None,256) the passed to dropout 3 which doesn't change the shape because it just drops the unit of neurons the connected to the final dense_2 layer which results in the shape of (None,8).

III Results and Discussions:

We had used 3 types of datasets, they are
1]Ravdness
2]Crema-D
3]Tess

S.No	Dataset Name	Accuracy %
1.	Ravdness	96%
2.	Tess	99%
3.	Crema-D	84%

Table: 3.1: Accuracy result for Dataset Ravdness, Crema-D, Tess

In the below Table:3.2 dataset outcomes, the results of three datasets over the emotions of every dataset and the accuracy in table :3.1, F1-score, support of every dataset are represented.

i. Ravdness: The below snapshots represent the results obtained while using the Ravdness dataset. The graphs plotted during Training and testing accuracy and loss are represented. An exactness of accuracy of 96% is obtained and the confusion matrix plotted using this dataset is also mentioned below.

ii. Tess: The below snapshots represent the results obtained while using the Tess dataset. The graphs plotted during Training and testing accuracy and loss are represented. An exactness of 99% is obtained and the confusion matrix plotted utilizing this dataset is likewise referenced underneath.

iii. Crema-D : The below snapshots represent the results obtained while using the Crema-D dataset. The graphs plotted during Training and testing accuracy and loss are represented. An accuracy of 84% is obtained and the confusion matrix in the figure 3.7 plotted using this dataset is also mentioned below.

-

Ravdness _Dataset				
	Precision	**Recall**	**F1-Score**	**Support**
Angry	0.94	0.96	0.95	141
Calm	0.96	1.00	0.98	129
Disgust	0.99	0.94	0.96	145
Fear	0.98	0.98	0.98	161
Happy	0.94	0.99	0.96	148

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Neutral	0.99	0.98	0.98	83
Sad	0.99	0.93	0.96	142
Surprise	0.95	0.95	0.95	131
Accuracy			0.96	1080
Macro avg	0.97	0.97	0.97	1080
Weighted avg	0.97	0.96	0.96	1080
Tess_Dataset				
	Precision	**Recall**	**F1-Score**	**Support**
Angry	0.99	1.00	0.99	282
Disgust	0.98	0.99	0.98	302
Fear	1.00	1.00	1.00	295
Happy	0.99	0.97	0.98	313
Neutral	1.00	1.00	1.00	314
Sad	1.00	1.00	1.00	301
Surprise	0.99	0.98	0.98	293
Accuracy			0.99	2100
Macro avg	0.99	0.99	0.99	2100
Weighted avg	0.99	0.99	0.99	2100
CREMA-D_Dataset				
	Precision	**Recall**	**F1-score**	**Support**
Angry	0.96	0.93	0.94	774
Disgust	0.77	0.84	0.80	726
Fear	0.88	0.76	0.82	748
Happy	0.87	0.86	0.86	812
Neutral	0.77	0.82	0.79	616
Sad	0.78	0.81	0.80	790
Accuracy			0.84	4466
Macro avg	0.84	0.84	0.84	4466
Weighted avg	0.84	0.84	0.84	4466

TABLE : 3.2: DATA SET OUTCOMES

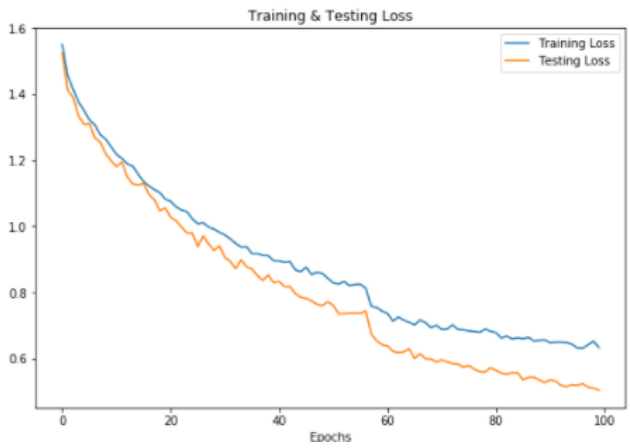

Fig 3.1: Ravdness_dataset test loss Graph

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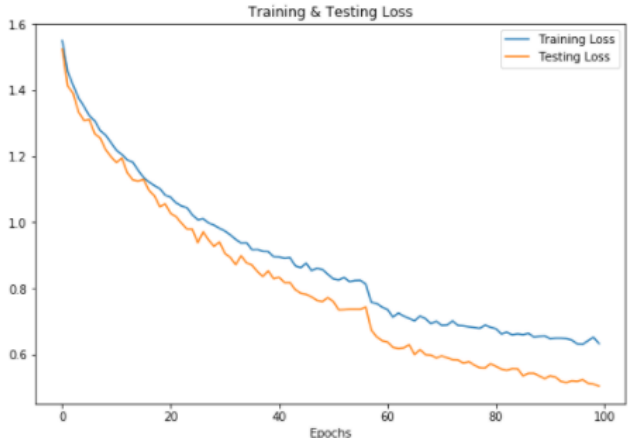

Fig 3.3: Tess_dataset test loss Graph

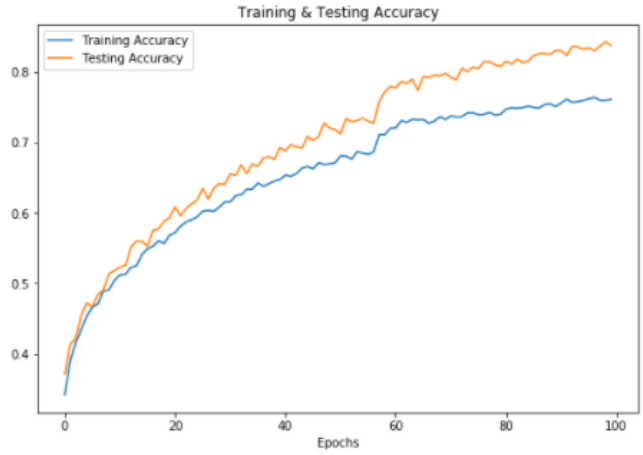

Fig 3.4: Tess_dataset test Accuracy Graph

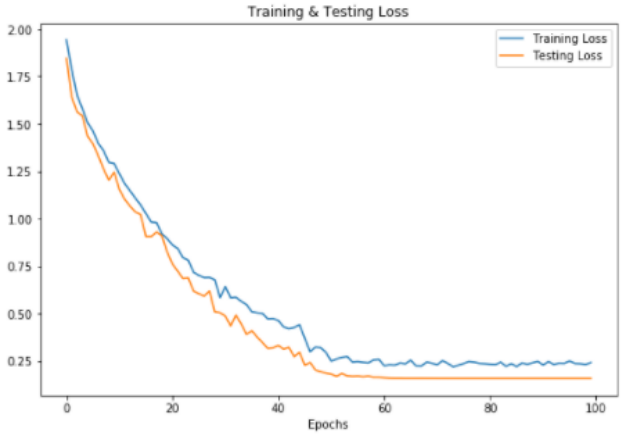

Fig 3.5: CREMA-D_dataset test loss Graph

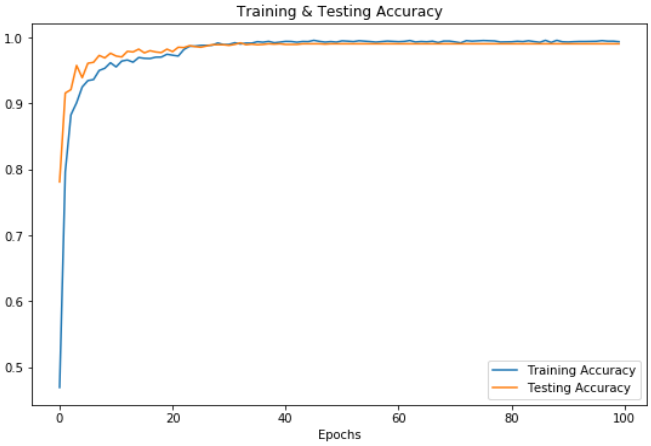

Fig 3.6: CREMA-D_dataset test Accuracy Graph

Fig: 3.2: Ravdness_dataset test Accuracy graph

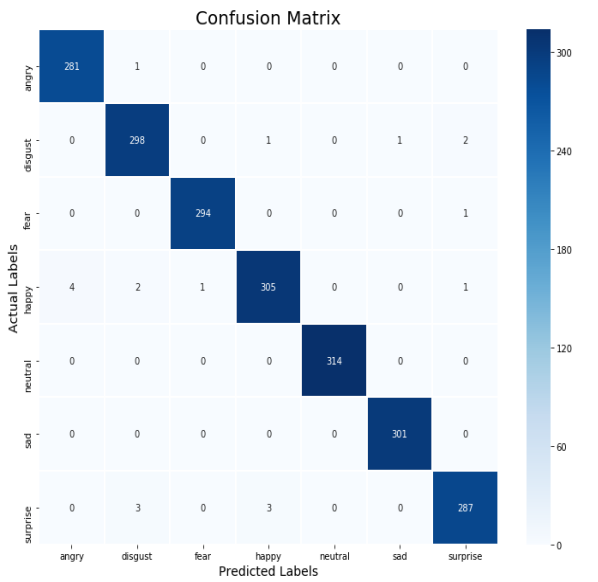

Fig 3.7: Confusion matrix

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IV Conclusion and Future Work

In this paper, emotions are detected using CNN (Convolution neural network) even in presence of noise. In future, these types of problems may be solved using other deep learning and natural language processing techniques. By building these types of models, it is efficient to connect with humans and their emotions and understand their feelings.

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